week 3 lab

Pass Task

1. What security-related configuration changes were performed on R3 by AutoSecure that were not performed in previous sections of the lab on R1?

    ```
    security authentication failure rate
    logging console critical
    enable password 7
    aaa new-model
    aaa authentication login local_auth local

    ### on all unused interfaces
    no ip redirects
    no ip unreachables
    no ip proxy-arp
    ip verify unicast source reachable-via rx 100

    ### on the internet port
    vrf forwarding Mgmt-intf
    no ip address
    no ip redirects
    no ip unreachables
    no ip proxy-arp
    negotiation auto
    no mop enabled
    ```

2. What security-related configuration changes were performed in previous sections of the lab that were not performed by AutoSecure?

    ```
    ### creating user01
    username user01 secret 9
    9$ImAUs9J44Hr7Aw$Ud08RPlGSo2unCBJYjmBcTvmNGxx4ZRfe/ux/wBQ0uM

    ip ssh authentication-retries 2

    ### all unused ports where shutdown
    ```

3. Identify at least five unneeded services that were locked down by AutoSecure and at least three security measures applied to each interface.
    1. Services disabled include:

        ```
        no ip bootp server
        no ip domain lookup
        no ip source-route
        ```

```
no ip gratuitous-arps
no license smart enable
```

2. For each interface, the following were disabled:
    1.
```
no ip redirects
no ip unreachables
no ip proxy-arp
ip verify unicast source reachable-via rx 100
no ip address
```

4. What are some advantages to using AutoSecure?
    1. assumes lacking technical knowledge so asks specific questions and assumes the worst, thus the program employees all available security options.
    2. relatively quick, can complete in under a minute
    3. prevents human error, all options on form are yes or no and the only thing someone can make a mistake on is the username and password

Credit task

1. Ping PC-C using the serial address of R1 as the source IP. Did it work? Explain why it did or did not work!
    1. No, Router 2's Serial interfaces have not been setup with OSPF RPA and will not carry traffic for router 1.

Distinction task

1. In the process issue the same test as with OSPF. Enable authentication on R1 and then try and ping PC-C. Did it
    1. No, like with OSPF R1 will refuse to communicate with its neighbors without the correct authentication protocol and a matching key.
    work? Explain
2. Besides “sh run” which show command tells me that authentication has been enabled in EIGRP?

Besides show run you can also use the following command to confirm if EIGRP authentication has been enabled

```
debug eigrp packets
```

if it is the debug messages will read.

```
*Aug 16 23:37:16.616: EIGRP: Sending HELLO on Se0/1/0 - paklen 60
*Aug 16 23:37:16.616:   AS 1, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0 iidbQ
```

```
un/rely 0/0
*Aug 16 23:37:17.472: EIGRP: received packet with MD5 authentication, key id =
1
*Aug 16 23:37:17.472: EIGRP: Received HELLO on Se0/1/0 - paklen 60 nbr
10.1.1.2
*Aug 16 23:37:17.472:   AS 1, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0 iidbQ
un/rely 0/0 peerQ un/rely 0/0
```

EIGRP evidence

```
### R1 ###

R1#sh ip eigrp nei
EIGRP-IPv4 Neighbors for AS(1)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q
Seq
                                                   (sec)         (ms)
Cnt Num
0   10.1.1.2                Se0/1/0                  12 00:10:29    2   100  0
17

R1#sh ip eigrp interfaces
EIGRP-IPv4 Interfaces for AS(1)
                            Xmit Queue   PeerQ        Mean   Pacing Time
Multicast    Pending
Interface             Peers  Un/Reliable  Un/Reliable  SRTT   Un/Reliable
Flow Timer   Routes
Se0/1/0                 1        0/0       0/0           2       0/12
50           0

### R2 ###

R2#sh ip eigrp nei
EIGRP-IPv4 Neighbors for AS(1)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q
Seq
                                                   (sec)         (ms)
Cnt Num
1   10.2.2.1                Se0/1/1                  14 00:10:41    3   100  0
11
0   10.1.1.1                Se0/1/0                  12 00:11:43    1   100  0
8

R2#sh ip eigrp interfaces
EIGRP-IPv4 Interfaces for AS(1)
                            Xmit Queue   PeerQ        Mean   Pacing Time
Multicast    Pending
Interface             Peers  Un/Reliable  Un/Reliable  SRTT   Un/Reliable
Flow Timer   Routes
```

```
Se0/1/0                   1        0/0       0/0           1       0/12
50           0
Se0/1/1                   1        0/0       0/0           3       0/12
50           0

### R3 ###

R3#sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q
Seq
                                                   (sec)         (ms)
Cnt Num
0   10.2.2.2                Se0/1/1                  14 00:11:15    2   100  0
16

R3#sh ip eigrp interfaces
EIGRP-IPv4 Interfaces for AS(1)
                              Xmit Queue   PeerQ        Mean   Pacing Time
Multicast    Pending
Interface             Peers   Un/Reliable  Un/Reliable  SRTT   Un/Reliable
Flow Timer   Routes
Se0/1/1                   1        0/0       0/0           2       0/12
50           0
```